

EE 587 Introduction to Robotics Homework #3

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1. Suppose that $a = (1, -1, 2)$ and that $R = R_{x,90}$. Show by direct calculation that you have $RS(a)R^T = S(Ra)$

Rotation Matrix $R_x(90^\circ)$ and Skew Symmetric Matrix S_a For $a = (1, -1, 2)$:

$$R_x(90^\circ) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$S(a) = \begin{bmatrix} 0 & -2 & -1 \\ 2 & 0 & -1 \\ 1 & 1 & 0 \end{bmatrix}$$

Calculating $RS(a)R^T$: Substituting R and S(a):

$$RS(a) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -2 & -1 \\ 2 & 0 & -1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$RS(a) = \begin{bmatrix} 0 & -2 & -1 \\ -1 & -1 & 0 \\ 2 & 0 & -1 \end{bmatrix}$$

$Ra =$

$$Ra = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ -1 \end{bmatrix}$$

Skew symmetric matrix of Ra :

$$S(Ra) = \begin{bmatrix} 0 & 1 & -2 \\ -1 & 0 & -1 \\ 2 & 1 & 0 \end{bmatrix} //$$

Then Calculation of $RS(a)R^T$:

$$RS(a)R^T = \begin{bmatrix} 0 & -2 & -1 \\ -1 & -1 & 0 \\ 2 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 & -2 \\ -1 & 0 & -1 \\ 2 & 1 & 0 \end{bmatrix} //$$

2. Given the Euler angle transformation $R = R_{z,\phi} R_{y,\theta} R_{x,\psi}$, show that

$$\frac{d}{dt} R = S(\omega) R \text{ Where}$$

$$\omega = (c_\psi s_\theta \dot{\Phi} - s_\psi \dot{\theta})i + (s_\psi s_\theta \dot{\psi} + c_\psi \dot{\theta})j + (\dot{\psi} + c_\theta \dot{\Phi})k$$

(the components of i, j and k are respectively called *nutation*, *spin* and *precession*)

For given Euler rotation general formula first finding the rotation matrices:

$$R = R_z(\phi) R_y(\theta) R_z(\psi)$$

$$R_z(\phi) = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_y(\theta) = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$$

$$R_z(\psi) = \begin{bmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Then we find derivation of R

Using the product rule:

$$\begin{aligned} \frac{dR}{dt} &= \frac{dR_z(\phi)}{dt} R_y(\theta) R_z(\psi) + R_z(\phi) \frac{dR_y(\theta)}{dt} R_z(\psi) \\ &\quad + R_z(\phi) R_y(\theta) \frac{dR_z(\psi)}{dt} \end{aligned}$$

Derivative of Each Component:

$$\frac{dR_z(\phi)}{dt} = \dot{\phi} \begin{bmatrix} -s\phi & -c\phi & 0 \\ c\phi & -s\phi & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\frac{dR_y(\theta)}{dt} = \dot{\theta} \begin{bmatrix} -s\theta & 0 & c\theta \\ 0 & 0 & 0 \\ -c\theta & 0 & -s\theta \end{bmatrix}$$

$$\frac{dR_z(\psi)}{dt} = \dot{\psi} \begin{bmatrix} -s\psi & -c\psi & 0 \\ c\psi & -s\psi & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Combining Angular Velocity Components:

$$\omega = \dot{\phi} \hat{z}_1 + \dot{\theta} \hat{y}_2 + \dot{\psi} \hat{z}_3$$

Skew-Symmetric Matrix for w :

$$S(w) = \begin{bmatrix} 0 & -w_z & w_y \\ w_z & 0 & -w_x \\ -w_y & w_x & 0 \end{bmatrix}$$

Relationship Between $\frac{dR}{dt}$ and $S(w)$:

$$\frac{dR}{dt} = S(w)R$$

Where $S(w)$ encodes the contributions of $\dot{\phi}$, $\dot{\theta}$, and $\dot{\psi}$ into a single matrix.

3. Let $R_{k,\theta}$ be the rotation matrix corresponding to a rotation around an axis defined by the vector $k = (k_1, k_2, k_3)$. Prove using the Rodrigues' Formula that we have

$$\frac{d}{d\theta} R_{k,\theta} = S(k)R_{k,\theta}$$

Problem statement and Rodrigues' rotation formula:

$$\frac{dR_{k,\theta}}{d\theta} = S(k)R_{k,\theta},$$

Rodrigue's Rotation

$$R_{k,\theta} = I + \sin\theta S(k) + (1 - \cos\theta) S(k)^2,$$

$$S(k) = \begin{bmatrix} 0 & -k_z & k_y \\ k_z & 0 & -k_x \\ -k_y & k_x & 0 \end{bmatrix}$$

Derivative of $R_{k,\theta}$ with respect to θ :

$$\frac{dR_{k,\theta}}{d\theta} = \frac{d}{d\theta} \left(\underset{\substack{\downarrow \\ O_{//}}}{I} + \sin\theta S(k) + (1 - \cos\theta) S(k)^2 \right)$$

$$\frac{d}{d\theta} (\sin\theta S(k)) = \cos\theta S(k)$$

$$\frac{d}{d\theta} ((1 - \cos\theta) S(k)^2) = \sin\theta S(k)^2$$

$$\frac{dR_{k,\theta}}{d\theta} = \cos\theta S(k) + \sin\theta S(k)^2$$

Rewriting Using $R_{k,\theta}$:

$$\overbrace{R_{k,\theta}}^{\leftarrow} = I + \sin\theta S(k) + (1 - \cos\theta) S(k)^2$$

$$\frac{dR_{k,\theta}}{d\theta} = S(k) (I + \sin\theta S(k) + (1 - \cos\theta) S(k)^2)$$

which simplifies to:

$$\frac{dR_{k,\theta}}{d\theta} = S(k) R_{k,\theta}$$

4. Two frames $\{0\}$ and $\{1\}$ are related by the Homogeneous transformation

$$H = \begin{bmatrix} 0 & -1 & 0 & 1 \\ 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

A particle has velocity $v_1(t) = (3, 1, 0)$ relative to frame $\{1\}$. Calculate the particle velocity relative to frame $\{0\}$?

$$T = \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix} \quad v_1(t) = (3, 1, 0)$$

Vel. in $\{0\}$, v_0 is related to its velocity in frame $\{1\}$, v_1 through the transformation:

$$v_0 = R \cdot v_1 + \dot{p}$$

$$R = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad v_1(t) = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} \quad p = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

$$v_0 = R \cdot v_1 + \dot{p}$$

$$\begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \\ 0 \end{bmatrix} //$$

Programming:

5. In this question, you will implement a simple “Robot controller” by using the *forward kinematics* and *inverse velocity kinematics* of a planar RR manipulator. We know all the kinematics and velocity kinematics equations of this planar robot from lecture notes. You can use either Matlab or Python to implement this question

a. Write code to “simulate” a Planar RR robot with link lengths $l_1 = 3$ and $l_2 = 2$ units.

Define the angular ranges for the joints as $\theta_1 = \{-90^\circ, 90^\circ\}$ and $\theta_2 = \{-170^\circ, 170^\circ\}$. For any given pair of joint variables, you should be able to plot the robot on a plane (different colors for each link? Small circles for the joints? A fat point for the end effector?)

b. Define two points in the robot workspace: The start point p_s and goal point p_g in the robot work-space. (You can later experiment with different start and end points)

c. We would like to implement a discrete-time proportional position controller in the work-space that would take the robot from the initial point to the goal point. For this, you will periodically (at each time step): (1) calculate the position error, (2) define a required end-effector velocity vector (your controller), (3) transform this into required joint velocities, (4) execute the joint motion to determine the new end-effector position, (5) repeat the cycle.

d. Experiment with different sampling periods and controller Proportional Gains.

e. Present your results in the report as “time snapshots” of the robot configuration and/or end effector positions. Also see (g) below.

f. You should be able to detect and raise an alarm if a singularity is encountered during the control of the robot.

g. Have a video recording (screen capture) of the execution of your controller for some example cases. Upload to YouTube as “unlisted” videos and share the link in your homework report.

```
In [ ]: import numpy as np
import scipy.linalg
from scipy.linalg import expm
import roboticstoolbox as rtb
from roboticstoolbox import *
from roboticstoolbox import ERobot2, Link2
from spatialmath import *
from spatialmath.base import *
from spatialgeometry import Sphere, Box
from swift import Swift
from spatialmath import SE3, SE2
from roboticstoolbox import quintic, trapezoidal, mttraj, mstraj, xplot, c
from mpl_toolkits.mplot3d import Axes3D
from roboticstoolbox.backends.PyPlot import PyPlot2, RobotPlot2
import math
import time
from math import pi
```

```

import matplotlib.pyplot as plt
from matplotlib import cm
from matplotlib.widgets import Slider
from matplotlib.animation import FuncAnimation
from matplotlib.patches import Rectangle, Circle
np.set_printoptions(linewidth=100, formatter={'float': lambda x: f"{x:8.4}"})

%matplotlib widget

```

```

In [ ]: # Define link lengths
l1 = 3 # Length of link1
l2 = 2 # Length of link2

# Robot simulation function
def plot_robot(q, l1, l2, ax):
    """
    Plot the planar RR robot in a given configuration.
    """
    # Forward kinematics
    theta1, theta2 = q
    x1, y1 = l1 * np.cos(theta1), l1 * np.sin(theta1) # End of link1
    x2, y2 = x1 + l2 * np.cos(theta1 + theta2), y1 + l2 * np.sin(theta1 + theta2)

    # Clear the axis
    ax.clear()
    ax.set_xlim(-5, 5)
    ax.set_ylim(-5, 5)
    ax.set_aspect('equal')
    ax.grid()

    # Plot link1 (blue rectangle)
    ax.add_patch(Rectangle((0, 0), l1, 0.2, angle=np.degrees(theta1), color='blue'))
    # Plot link2 (red rectangle)
    ax.add_patch(Rectangle((x1, y1), l2, 0.2, angle=np.degrees(theta1 + theta2), color='red'))

    # Plot joints and end-effector
    ax.plot([0, x1, x2], [0, y1, y2], '-o', markersize=8, color='black')
    ax.add_patch(Circle((x2, y2), radius=0.1, color='orange')) # End-effector
    ax.set_title("Planar RR Robot")
    ax.set_xlabel("X (m)")
    ax.set_ylabel("Y (m)")

# Test the robot plot
fig, ax = plt.subplots()
q_test = [np.deg2rad(45), np.deg2rad(-45)] # Joint angles in radians
plot_robot(q_test, l1, l2, ax)
plt.show()

```

```

In [ ]: # Start and goal points in the workspace
p_s = np.array([3, 2]) # Start point
p_g = np.array([4, -1]) # Goal point

# Print workspace points
print("Start Point (p_s):", p_s)
print("Goal Point (p_g):", p_g)

```

```

In [ ]: def proportional_controller(l1, l2, p_s, p_g, Kp=1.0, dt=0.1, max_steps=1000):
    """
    Proportional position controller for a planar RR robot.
    """

```

```

"""
def inverse_kinematics(x, y, l1, l2):
    c2 = (x**2 + y**2 - l1**2 - l2**2) / (2 * l1 * l2)
    if abs(c2) > 1:
        raise ValueError("Target point is outside the workspace!")
    s2 = np.sqrt(1 - c2**2)
    theta2 = np.arctan2(s2, c2)
    theta1 = np.arctan2(y, x) - np.arctan2(l2 * s2, l1 + l2 * c2)
    return [theta1, theta2]

# Initial configuration
q = inverse_kinematics(p_s[0], p_s[1], l1, l2)

# Store trajectory
trajectory = [list(q)] # Ensure the trajectory stores lists

for step in range(max_steps):
    # Forward kinematics to compute current end-effector position
    theta1, theta2 = q
    x_current = l1 * np.cos(theta1) + l2 * np.cos(theta1 + theta2)
    y_current = l1 * np.sin(theta1) + l2 * np.sin(theta1 + theta2)
    p_current = np.array([x_current, y_current])

    # Calculate position error
    error = p_g - p_current

    # Check for convergence
    if np.linalg.norm(error) < 1e-2:
        print(f"Goal reached in {step} steps!")
        break

    # Compute end-effector velocity
    v = Kp * error

    # Compute Jacobian
    J = np.array([
        [-l1 * np.sin(theta1) - l2 * np.sin(theta1 + theta2), -l2 * np.
        [l1 * np.cos(theta1) + l2 * np.cos(theta1 + theta2), l2 * np.
    ])
    J_inv = np.linalg.pinv(J) # Pseudo-inverse for safety

    # Compute joint velocities
    q_dot = J_inv @ v

    # Update joint positions
    q = q + q_dot * dt
    trajectory.append(list(q)) # Append the current joint configurat

return trajectory

```

```

In [ ]: trajectory = proportional_controller(l1, l2, p_s, p_g, Kp=0.5, dt=0.1, ma
# Initialize the figure and axes
fig, ax = plt.subplots()
ax.set_xlim(-5, 5)
ax.set_ylim(-5, 5)
ax.set_aspect('equal')
ax.grid()

ax.plot(p_s[0], p_s[1], 'go', label='Start Point') # Start point (green
ax.plot(p_g[0], p_g[1], 'ro', label='Goal Point') # Goal point (red circ

```

```

# Add elements for the links and the end-effector
link1_line, = ax.plot([], [], '-o', color='blue', label='Link 1') # Link
link2_line, = ax.plot([], [], '-o', color='red', label='Link 2') # Link
end_effector_circle = ax.plot([], [], 'o', color='orange', label='End-Eff

trace_x, trace_y = [], [] # To store the end-effector path
trace, = ax.plot([], [], '--', color='orange', label='End-Effector Path')

def is_singular(J):
    """
    Detect singular configurations based on Jacobian condition number.
    """
    cond = np.linalg.cond(J)
    return cond > 1e3 # Example threshold for singularity

# Animation update function
def update_plot(frame):
    """
    Update the plot for the current frame of the animation.
    """
    # Extract joint angles for the current frame
    q = trajectory[frame] # Get joint angles for the current frame
    theta1, theta2 = q

    # Forward kinematics
    x1, y1 = l1 * np.cos(theta1), l1 * np.sin(theta1) # End of link1
    x2, y2 = x1 + l2 * np.cos(theta1 + theta2), y1 + l2 * np.sin(theta1 +

    # Compute the Jacobian
    J = np.array([
        [-l1 * np.sin(theta1) - l2 * np.sin(theta1 + theta2), -l2 * np.si
        [l1 * np.cos(theta1) + l2 * np.cos(theta1 + theta2), l2 * np.cos
    ])

    # Check for singularity
    if is_singular(J):
        end_effector_circle.set_color('purple') # Highlight singularitie
    else:
        end_effector_circle.set_color('orange') # Reset to orange

    # Update link1
    link1_line.set_data([0, x1], [0, y1]) # Base to end of link1

    # Update link2
    link2_line.set_data([x1, x2], [y1, y2]) # End of link1 to end of lin

    # Update end-effector
    end_effector_circle.set_data([x2], [y2]) # Position of end-effector

    # Update the trace
    trace_x.append(x2)
    trace_y.append(y2)
    trace.set_data(trace_x, trace_y)

# Create the animation
ani = FuncAnimation(fig, update_plot, frames=len(trajectory), interval=10

plt.legend()
plt.show()

```

```
In [ ]: # Create sliders for Kp and dt
ax_kp = plt.axes([0.25, 0.01, 0.65, 0.03]) # Position of Kp slider
ax_dt = plt.axes([0.25, 0.05, 0.65, 0.03]) # Position of dt slider

slider_kp = Slider(ax_kp, 'Kp', 0.1, 2.0, valinit=0.5)
slider_dt = Slider(ax_dt, 'dt', 0.01, 0.2, valinit=0.1)

def update_controller(val):
    """
    Update the controller with new Kp and dt values from sliders.
    """
    global trajectory
    Kp = slider_kp.val
    dt = slider_dt.val
    trajectory = proportional_controller(l1, l2, p_s, p_g, Kp=Kp, dt=dt,
    ani.event_source.stop()
    ani.event_source.start()

slider_kp.on_changed(update_controller)
slider_dt.on_changed(update_controller)
```

```
In [ ]: # Convert trajectory to radians for plotting
trajectory = np.array(trajectory)

# Plot joint trajectories
plt.figure()
for i in range(trajectory.shape[1]):
    plt.plot(trajectory[:, i], label=f'Joint {i + 1}')
plt.legend()
plt.xlabel('Point Index')
plt.ylabel('Joint Angle (rad)')
plt.title('Joint Trajectories')
plt.grid()
plt.show()
```

6. Review the P. Corke book and *Robotics Toolbox for Python* documentation to list (with short explanations) fundamental functions that calculate the *forward* and *inverse velocity kinematics* of robot arms as well as the *inverse position kinematics*. You may illustrate some examples that use these functions.

Programming Tasks Planar RR Manipulator Simulation (Question 5)

Steps:

Robot Simulation (Part a):

Create a visualization for the planar RR robot in Python/Matlab.

Use link lengths $l_1=3$ and $l_2=2$, plotting the links and joints for a given (θ_1, θ_2) .

Workspace Definition (Part b):

Define start psp and goal ppg positions in Cartesian coordinates within the workspace.

Discrete-Time Controller (Part c):

Implement a proportional controller:

- Compute the position error $e = p_g - p_{current}$.
- Calculate end-effector velocity: $v_{ee} = K_p \cdot e$ (adjust K_p experimentally).
- Use the Jacobian to compute joint velocities: $\dot{q} = J^{-1} \cdot v_{ee}$.
- Update joint positions $q = q + \dot{q} \cdot dt$.
- Plot the robot configuration and end-effector trajectory.

Experimentation (Parts d-g):

- Vary the proportional gain K_p and time step dt .
- Visualize snapshots of the robot moving toward the goal.
- Handle singularity detection by checking the determinant of the Jacobian matrix.

Optional:

- Record and upload a video of the controller executing for different cases.

Functions in P. Corke's Toolbox (Question 6)

Explore Python's Robotics Toolbox or Matlab equivalent. Identify and list functions for:

- Forward kinematics.
- Inverse kinematics.
- Velocity transformations.

Provide short explanations and include examples.